DONGSOO LEE, M.P.H.

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PROFESSIONAL SUMMARY

Ph.D. candidate in Biostatistics & Bioinformatics with 3+ years of industry R&D experience designing, analyzing, and reporting **clinical validation studies** for regulated medical products. Track record from study design and statistical analysis through **FDA 510(k) clearance**, including authoring validation reports and presenting statistical findings to regulatory and non-statistical audiences. Seeking a statistician role applying statistical and machine-learning methods to drug target discovery and clinical trials.

CORE COMPETENCIES

- **Statistical Methods:** Statistical inference, hypothesis testing and multiplicity control, simulation-based evaluation of type I error and power, statistical genetics (GWAS, TWAS, eQTL, fine-mapping, co-localization), machine learning
- **Genomics & Multi-omics:** Single-cell RNA-seq & ATAC-seq, multiome integration, bulk RNA-seq, cell-type deconvolution, multi-ancestry analysis
- **Clinical & Regulatory:** Clinical validation study design and analysis, statistical reporting for regulatory submission, FDA 510(k) clearance, medical imaging analysis, cross-functional R&D leadership
- **Programming:** SAS (Base/Advanced certified), R, Python, SQL, SPSS
- **Tools & Infrastructure:** Microsoft Office (Word, Excel, PowerPoint), Docker, Git, High-Performance Computing (SLURM)
- **Certifications:** SAS Certified Advanced Programmer (SAS 9); SAS Certified Base Programmer (SAS 9)
- **Languages:** Korean (native); English (full professional proficiency)

PROFESSIONAL & RESEARCH EXPERIENCE

Emory University - Atlanta, GA | Sep 2023 - Present

Graduate Researcher, Biostatistics & Bioinformatics (Ph.D. Candidate)

- Developing **transfer-learning statistical frameworks** for high-dimensional genomic association studies, integrating single-cell and bulk RNA-seq to nominate Alzheimer's disease risk genes for therapeutic follow-up
- Designed statistical methods to map **cell-type-specific enhancer-target gene regulatory** interactions from single-cell multimodal data
- Integrated scRNA-seq and scATAC-seq to characterize toxicant-induced disruption of developmental gene-regulatory programs, informing mechanisms of environmental disease risk

VUNO Inc. - Seoul, South Korea | Feb 2020 - Aug 2023

Research Scientist, Brain Team (Research Lead, 2022–2023)

- Authored clinical validation reports and led cross-functional teams to secure **FDA 510(k) clearance** for an AI-powered brain MRI segmentation product (VUNO Med-DeepBrain) supporting Alzheimer's diagnosis
- Developed and validated a deep-learning model fusing imaging and tabular clinical features to predict medial temporal lobe atrophy (MTA) scores at **>80% accuracy**; deployed as a commercial product feature
- Led the EUROSTAR DEPREDICT consortium on MRI-based prediction of major depressive disorder, coordinating multi-site analysis across 3 international institutions (Amsterdam UMC, University of Oslo, NNL)
- Inventor on **3 granted U.S. patents** in medical imaging and disease prediction (first and co-inventor)

EDUCATION

Emory University, Rollins School of Public Health - Atlanta, GA

Ph.D. Candidate, Biostatistics & Bioinformatics | Expected 2028

M.S., Biostatistics | 2026

Seoul National University - Seoul, South Korea | 2020

M.P.H., Biostatistics

Thesis: Genome-wide assessment of MRI traits implicates *SHARPIN* in Alzheimer pathogenesis

Konkuk University - Seoul, South Korea 2018
B.A., Applied Statistics

SELECTED PUBLICATIONS

**equal contribution; †corresponding author. Total 10 peer-reviewed articles.*

- Su C†, **Lee D**, Jin P, Zhang J† (2025). Cell-type-specific mapping of enhancers and target genes from single-cell multimodal data. *Nature Communications*, 16(1), 3941.
- Lee SJ*, **Lee D***, Suh CH†, Jeong SY, Shin HM, Jung W, et al. (2025). Development and validation of a deep learning–based automatic classification algorithm for the medial temporal lobe atrophy score using a multi-modality cascade transformer. *Clinical Radiology*, 106993.
- Park JY, Lee JJ, Lee Y, **Lee D**, Gim J, Farrer L, Lee KH†, Won S† (2023). Machine learning–based quantification for disease uncertainty increases the statistical power of genetic association studies. *Bioinformatics*, 39(9), btad534.
- **Lee D**, Suh CH†, Kim J, Jung W, Park C, Jung KH, et al. (2022). Augmenting MRI with tabular features for enhanced and interpretable medial temporal lobe atrophy prediction. *MICCAI Workshop on Machine Learning in Clinical Neuroimaging* (pp. 125–134). Springer.
- Park JY, **Lee D**, Lee JJ, Gim J, Gunasekaran TI, Choi KY, et al., Lee KH† (2021). A missense variant in *SHARPIN* mediates Alzheimer's disease–specific brain damage. *Translational Psychiatry*, 11(1).
- Kim J*, **Lee D***, Woo H, Kim H, Son B, Park M, Hong S† (2020). Impact of particulate matter on the clinical characteristics of rhinitis. *The Laryngoscope*.

PATENTS

3 granted U.S. patents; 10+ international patents.

- **Lee DS**, Oh H, Park S, Sung J, Hong E, Kim WJ, et al. Method for detecting white matter lesions based on medical image. U.S. Patent 12,236,605 (Feb 25, 2025).
- Oh H, Park S, Sung J, Kim WJ, Hong E, **Lee DS**. Method for predicting disease based on medical image. U.S. Patent 12,125,199 (Oct 22, 2024).
- Hong E, Jung W, Park S, Oh H, **Lee DS**, Kim WJ, Sung J. Estimating cortical region thickness by extracting interfaces as mesh data. U.S. Patent 12,094,147 (Sep 17, 2024).

HONORS & AWARDS

- **Patel–Naik Award** (1st place, research proposal), Emory University, 2026
- ENCORE (Emory Network of Computational Omics Research) Lab Manager, 2025-2026
- Cole and Bain Scholarship, Emory University, 2023
- Brain Korea 21 Plus Scholarship, Seoul National University, 2018
- Third Prize, Poster Session, KOSHIS Fall Conference, 2019
- **Graduate Teaching Assistant**, Emory University: Introduction to Data Science I (DATA515, 2026), Probability I (BIOS 512, 2026), Data Science Toolkit (DATA 550, 2025), Statistical Inference I (BIOS 513, 2025), SAS Programming (BIOS 531, 2024)

SELECTED PRESENTATIONS

- **IGES**, October 2026 (platform presentation). *TL-TWAS: Transfer Learning for TWAS Using Large-Scale Transcriptomic Resources in Underpowered Target Contexts.*
- **STATGEN**, May 2026 (invited session). *scTWASTransfer: single-cell transcriptome-wide association studies via transfer learning with bulk and single-cell data.*
- **RSNA Annual Meeting**, Nov 2022. *Deep learning–based automatic MTA score classification using a multi-modality cascade transformer.*
- **MICCAI Workshop on Machine Learning in Clinical Neuroimaging**, Sep 2022. *Augmenting MRI with tabular features for interpretable MTA prediction.*